

Challenge (Habanero)

- Q1.** A construction company mixes 2 parts of cement with 3 parts of sand to make concrete. If they need 50 bags of concrete, how many bags of cement will they need?
- (A) 20
(B) 25
(C) 30
(D) 35
(E) 40
- Q2.** A chef mixes 1 part of sugar with 2 parts of flour to make a cake. If she has 12 cups of sugar, how many cups of flour will she need?
- (A) 12
(B) 16
(C) 20
(D) 24
(E) 28
- Q3.** A builder mixes 3 parts of water with 1 part of paint to make a paint mixture. If he needs 14 gallons of paint mixture, how many gallons of water will he need?
- (A) 10
(B) 12
(C) 10.5
(D) 12.5
(E) 11.2
- Q4.** A recipe calls for a mixture of 1 part of oil with 4 parts of vinegar. If 5 cups of oil are needed, how many cups of vinegar are required?
- (A) 10
(B) 15
(C) 20
(D) 25
(E) 30
- Q5.** A construction site needs a mixture of 2 parts of gravel with 1 part of cement. If 30 tons of gravel are available, how many tons of cement will they need?
- (A) 10
(B) 15
(C) 12
(D) 18
(E) 20
- Q6.** A chef makes a spice blend using 3 parts of salt and 2 parts of pepper. If 24 cups of salt are needed, how many cups of pepper will be required?
- (A) 12
(B) 16
(C) 14
(D) 20
(E) 18
- Q7.** A builder needs to mix 1 part of glue with 5 parts of water. If 6 gallons of glue are needed, how many gallons of water will they require?
- (A) 20
(B) 25
(C) 30
(D) 35
(E) 40
- Q8.** A chef mixes 4 parts of water with 1 part of food coloring. If 8 cups of water are needed, how many cups of food coloring will they need?
- (A) 1
(B) 2
(C) 4
(D) 6
(E) 8

Open Ended Questions (FRQ)

- Q9.** A construction company needs to make a mixture of 2 parts of sand and 1 part of cement. If they have 18 bags of cement, how many bags of sand do they need to make the mixture? Show your work and check your units.
- Q10.** A chef needs to make 50 pounds of a spice blend using 3 parts of paprika and 2 parts of garlic powder. If they have 20 pounds of paprika, how many pounds of garlic powder do they need? Show your work and check your units.

Chef's Tips

- Read each question carefully
- Check your units
- Use the space provided to show your work